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01 Introduction HKA: Integration of Digital Twins (AAS) and CAD (Siemens NX)

Within the scope of this development project, the informational and conceptual bridge between the Asset Administration Shell (AAS) and the CAD system Siemens NX is built.

> "The Asset Administration Shell is the standardized digital representation of an asset. It structures all relevant data and properties of a physical or logical object throughout its entire lifecycle, thus forming the technical foundation for the Digital Twin in Industry 4.0." [1]

The goal is to enable working with Digital Twins across system boundaries and to ensure continuous data utilization throughout the entire product lifecycle.

Methodologically, the project is divided into two consecutive phases:

Phase 1: Fundamentals and Application Scenarios in the Product Lifecycle

In the first phase of the project, the theoretical fundamentals of the AAS were developed, and the specific added value and application scenarios offered by connecting a CAD system to the administration shell were evaluated.

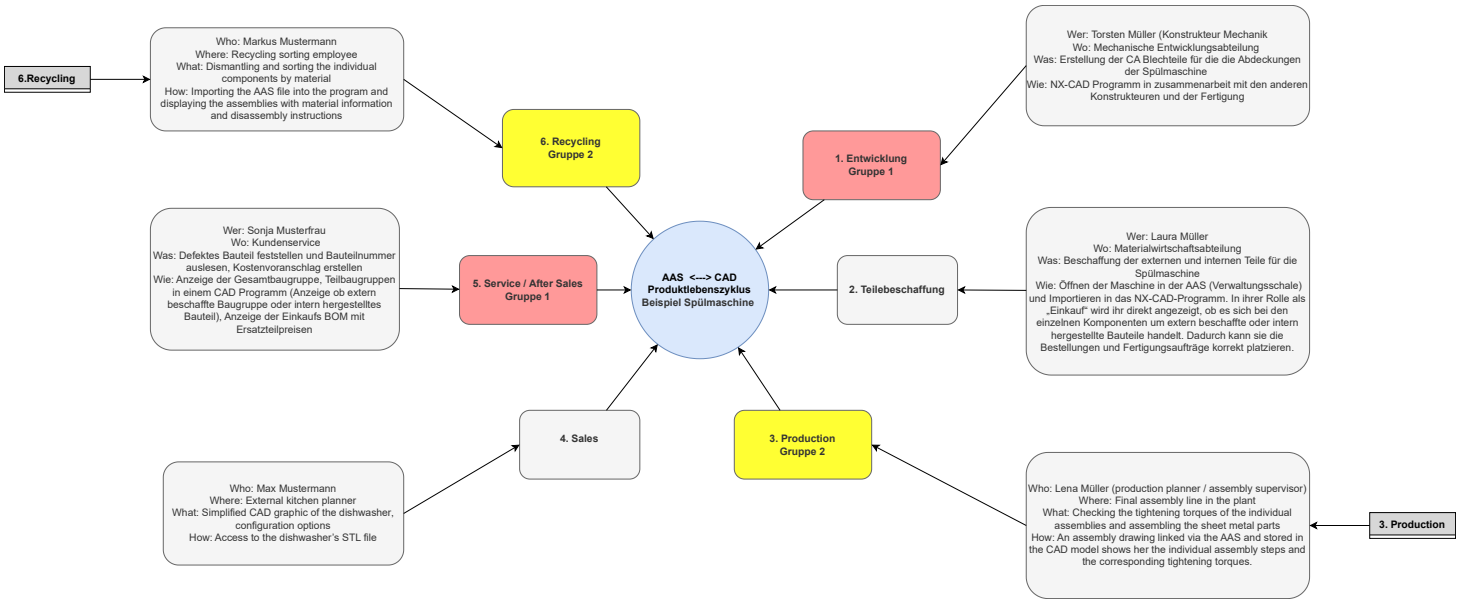
A practical example of a **dishwasher** serves to illustrate the concepts. Along its product lifecycle, highly varying requirements arise in each phase regarding the provision, updating, and structuring of data from the Digital Twin.

The considered lifecycle encompasses the following six phases:

1. Development (CAD & Engineering)
2. Parts Procurement
3. **Production (Manufacturing)**
4. Sales
5. Service / After-Sales

6. **Recycling (End-of-Life)**

*As shown in the following diagram, these phases form a continuous cycle. Within the scope of this project, the **Production** and **Recycling** phases, in particular, were elaborated in depth. The detailed elaboration of these use cases is the subject of the following chapters.*



Phase 2: Technical Implementation and Interface Functions

Based on the conceptual preliminary work of the first phase, the focus of the second project phase was on the technical implementation. Here, the central core functions required for practical work with AASX data packages and the CAD program Siemens NX were identified and developed.

> **Documentation Note ("Living Document"):**
> Since research and work on this repository will continue in the future, this documentation is designed directly as the structure for the GitHub repository (`README.md` and linked Markdown files). Redundant, separately maintained documentation was deliberately avoided. This ensures that code changes and documentation always remain synchronous, transparently versioned, and accessible in a central location for future developers.

3. Scenario: Production (Manufacturing)

3.1 Detailed description of the scenario

Lena Müller works as a production planner and assembly manager in final assembly. She is responsible for the correct execution of the assembly processes as well as for the quality assurance of the assembled modules. Lena checks the tightening torques of the individual screw connections and assembles the corresponding sheet metal components. In doing so, she ensures that all assembly specifications are strictly adhered to in order to guarantee the functionality, safety, and quality of the product.

The relevant assembly information is provided to Lena digitally. An assembly drawing, linked to the respective physical asset via the Asset Administration Shell (AAS), is stored in the CAD model. Using a digital device (e.g., an industrial tablet or HMI), Lena can access this information in context.

3.2 Process flow description

- **Product / Asset Identification:** Lena identifies the product to be assembled using a serial number or a QR code.
- **Calling up the AAS:** The corresponding Asset Administration Shell is loaded via the manufacturing system.
- **Accessing assembly information:** The AAS points to the linked CAD model and the corresponding assembly drawing.
- **Display of assembly steps:** The individual assembly steps are presented sequentially, including the required tools.
- **Checking tightening torques:** The target torques for each screw connection are displayed and verified with the torque tool in use.
- **Assembly of sheet metal components:** Lena assembles the sheet metal components according to the drawing and the defined process steps.
- **Documentation and feedback:** The completed assembly and the verified torques are documented in the system and, if necessary, reported back to higher-level systems.

3.3 Required files and resources

- **CAD models** (e.g., STEP, JT, native CAD formats)
- **Assembly drawings** (2D/3D, e.g., PDF or directly in CAD)
- **Torque specifications** (e.g., tables or metadata)
- **Asset Administration Shell** (AAS model in JSON or XML format)
- **Digital devices** (tablet, industrial PC, HMI)

- **Torque tools** (ideally with a digital interface)
- **MES/PLM systems** for data management

3.4 Technical implementation

3.4.1 Implementation in the Asset Administration Shell (AAS)

The Asset Administration Shell serves as a central data hub for assembly- and quality-relevant information. Technically, the implementation is carried out via clearly defined `SubmodelElements` that standardize assembly knowledge, CAD references, and torque specifications.

Assembly and torque data

- **Submodel "AssemblyInformation":** Contains the complete description of the assembly steps as structured `SubmodelElements` (e.g., `Operation`, `Sequence`, `StepNumber`).
- **Tightening torques:** Each screw connection is described by a `Property` element:
  - `FastenerID`
  - `TorqueNominal` (e.g., 12 Nm)
  - `TorqueTolerance` (e.g., ±10%)
  - `ToolType` (e.g., electronic torque wrench)

> **Example:** The screw connection of the side panel has the attribute `TorqueNominal = 12 Nm`, `TorqueTolerance = ±1.2 Nm`.

3.4.2 Implementation in the CAD environment

The CAD environment (viewer or assembly frontend at Lena's workstation) acts as a visual and interactive interface between the worker and the digital product twin.

- Each component and screw connection in the CAD model is linked to the `InstanceId` of the corresponding AAS component.
- If Lena clicks on a screw in the 3D model, the **target torque**, **tolerance**, and **tool type** are loaded directly from the AAS and displayed.

Feedback and reporting channel (As-Planned vs. As-Built)

Lena can report deviations directly in the CAD interface, e.g.:

- Screw is inaccessible
- Component does not fit stress-free
- Torque cannot be achieved structurally

These reports are stored as an `Event` or `Annotation` directly in the AAS and are made available to engineering and production planning.

6. Scenario: Recycling (End-of-Life)

6.1 Detailed description of the scenario

In 2026, the linear economy ("take, make, dispose") was legally ended by the *Ecodesign for Sustainable Products Regulation* (ESPR). Every dishwasher sold in the EU must now have a Digital Product Passport (DPP), which is technically implemented as an Asset Administration Shell (AAS).

Markus Mustermann, a specialized employee in a certified dismantling center, faces a challenge: A modern dishwasher consists of over 150 different materials, including valuable stainless steel, copper-bearing motors, electronics with rare earths, and problematic bitumen insulation mats.

Without the AAS, the dishwasher would be a "black box" for Markus. He would have to shred it, which severely degrades material purity. Thanks to the AAS, Markus becomes an Urban Miner: He accesses the digital twin, which tells him not only what is installed but also how to separate it non-destructively to maximize recycling yields.

6.2 Process flow description

The workflow is divided into five phases:

1. **Capture & Initialization:** Markus scans the QR code lasered onto the housing frame. The AAS opens the corresponding model for this.
2. **Status & Safety Analysis:** Before Markus uses any tools, the system checks the "Safety" submodel. It displays whether capacitors need to be discharged or if chemical residues (e.g., rinse aid remnants) are present in the hoses.
3. **Virtual Exploration:** Markus receives the dismantling drawings in the AAS and disassembles the machine based on them. He is shown which part is made of which material. At the end, he enters how much mass of each material he has separated by type.
4. **Guided Dismantling:** Markus follows the steps stored in the AAS. The system dictates:
    - *Step 1:* Loosen the Torx screws (T15) on the back panel.
    - *Step 2:* Disconnect the wiring harness.
    - *Step 3:* Remove the insulation.

6.3 Required files and resources

The following items are required for the implementation:

- **The AASX container file:** Contains the logical structure and all references.
- **Identification:** Unique ID.
- **Digital Product Passport (DPP):** Contains the legally required ESPR data (carbon footprint, recyclate content).
- **Bill of Materials (R-BOM):** The hierarchical list as well as properties for recycling all components.
- **TechnicalProperties:** Material data sheets per component, provided by the respective manufacturers and merged into the dishwasher's AAS.
- **CAD resources:** Dismantling drawings and 3D model for disassembly.

6.4 Technical implementation

6.4.1 Implementation in the Asset Administration Shell (AAS)

The AAS serves as a data hub. All installed components have their properties stored by the respective manufacturers, and these are assembled into a dishwasher.

- **Material Identification:** Every component in the dishwasher receives a `Property` element with the material ID.
  - *Example:* The cladding elements receive the attribute `Material: V2A (Stainless Steel)`.
- **API Interface:** The AAS provides a REST API through which the recycling software makes queries (e.g., `GET /submodels/MaterialComposition/submodel-elements/CopperWeight`).

6.4.2 Implementation in the CAD environment

The CAD environment serves as the central data source for creating the Asset Administration Shell (AAS). The process is structured as follows:

- **Provision of documents:** Dismantling drawings in PDF format are generated directly from the CAD system and made available to the AAS.
- **Material flow analysis:** An automated list is generated that records all components, including their volumes and materials. From this, the exact masses of the different raw materials can be derived.
- **Target-actual comparison in the recycling process:** As part of the recycling process, this data is calculated for each dishwasher variant and stored in the AAS as target values.
- **Proof of obligation:** After dismantling, the recycler is obliged to enter the actually recovered masses of the different materials as actual values into the AAS.

- **Feedback loop to the manufacturer:** This data is fed back to the manufacturer. This gives the manufacturer seamless proof of the percentage to which the machine was sorted and dismantled.
- **Visualization:** Given high system complexity, an additional 3D model can be provided to the recycler.
- **Additional visualization:** The system calculates the recycling priority based on the stored material properties. Components containing pollutants are displayed flashing red.

6.5 Regulations and Legal Guidelines (Additional Requirements)

1. **ESPR (Ecodesign for Sustainable Products Regulation)**
    - **Standard tools & dismantlability:** The existing Regulation (EU) 2019/2022 for dishwashers already stipulates that certain parts (such as pumps or motors) must be replaceable with "commonly available tools" and without causing permanent damage to the appliance.
    - **Dismantling time:** The new ESPR (Regulation (EU) 2024/1781) allows the EU to set explicit performance requirements in product-specific "delegated acts". For the revision of the dishwasher rules (planned from 2026), discussions are underway to include dismantling time as a metric for recyclability. [2], [3]
2. **EU Battery Regulation (2023/1542)**
    - **Removability:** From February 18, 2027, portable batteries must be designed so that they can be easily removed and replaced by the end user or qualified personnel.
    - **Battery passport:** While the full "battery passport" primarily applies to industrial (>2 kWh) and traction batteries, information on chemical composition and safe removal for small batteries must be stored in the Digital Product Passport (DPP) of the main device (here, the dishwasher's AAS). [4]
3. **WEEE Directive (Waste Electrical and Electronic Equipment)**
    - WEEE regulates the handling of electronic waste after use.
    - **Collection and recovery targets:** For large appliances (Category 1), which include dishwashers, the directive prescribes a recovery target of 85% and a target for preparation for reuse and recycling of 80%.
    - **Role of the AAS:** The AAS is the tool to guarantee "material hygiene". These high targets can only be achieved economically through single-origin separation (e.g., grade 1.4301 stainless steel instead of mixed scrap). [5]
4. **Substances of Concern (REACH & SCIP)**
    - **Information obligation:** Since January 2021, manufacturers must submit information on Substances of Very High Concern (SVHC) to the ECHA SCIP database if the concentration exceeds 0.1%.

- **AAS Integration:** The AAS serves as an interface to make this complex chemical data directly visualizable for Markus Mustermann at his workstation (e.g., warning label for bitumen mats or flame retardants in circuit boards), without him having to search manually in databases. [6], [7]

02 Introduction GitHub

AAS-CAD Integration: Connection between Asset Administration Shell and CAD

Introductory document for our AAS-CAD Integration project.

In this project, we are working on enabling the Asset Administration Shell (AAS) – the standardized digital twin – to communicate directly with our CAD systems. The goal is to enable bidirectional data exchange and to provide concrete code examples and workflows for this purpose.

What exactly is this about?

The current state of the art is: 3D models reside in CAD, and the digital twin lives completely isolated from it in its own software environment. However, if we want to implement use cases like intelligent recycling or seamless material tracking, these two worlds must talk to each other. This repository collects scripts and plug-ins to build exactly this bridge.

Core Features Overview

We focus on four fundamental functions to establish the interaction between AAS and CAD. In the table, you will find direct links to the respective explanations and implementations for Siemens NX and

PTC Creo, as well as the current development status for Autodesk Fusion 360.

#	Feature	Implementation in Siemens NX	Implementation in PTC Creo	Implementation in Autodesk Fusion 360
1	**Import of 3D data from the AAS** *(Loading the model from the digital twin into CAD)*	NX Part Loader	Creo Doc	🚧 Work in Progress
2	**Assignment of an Asset ID to a part** *(The fundamental link between the 3D model and the AAS instance)*	NX Asset-ID assign	Creo Doc	🚧 Work in Progress
3	**Synchronize properties** *(Exchanging material data, metadata, etc.)*	NX Sync Properties with AAS	Creo Doc	🚧 Work in Progress
4	**Data export from CAD** *(currently as CSV)*	NX to AAS Export	Creo Doc	🚧 Work in Progress

Detailed Description of the Features

How exactly these four functions work in detail and how they are solved on the system side can be found directly in the linked documents from the table above. There you will find detailed explanations and code examples for Siemens NX and PTC Creo.

Contributing

We always welcome support. If you want to contribute your own scripts, have a solution for another CAD system like SolidWorks, or find bugs, feel free to create a Pull Request.

License

The core project "Connecting CAD to Digital Twins" is licensed under the **Apache License 2.0**. Detailed information can be found in the LICENSE.md file.

> **Important note regarding Siemens NX (UI-Styler):** > This project contains UI code generated with the **Siemens NX 2021 UI-Styler** and manually adapted. To compile or run this code, proprietary NX Open libraries (DLLs) are required, which are **not** part of this repository. The rights to these libraries and to the basic framework of the generated code remain with Siemens AG. Please observe the corresponding license terms (EULA) of your Siemens NX installation.

Further legal notices are listed in the NOTICE.md file.

NX AASX to STEP Import Tool

This project contains a Python script that is executed via the **NX Open Python API**.
The tool is used to select an **AASX file (Asset Administration Shell)**, automatically extract a contained **STEP model** from it, and then import it into **Siemens NX**.

The application thus connects an **Asset Administration Shell (AAS)** with a CAD system by directly opening a STEP model referenced within the AAS in NX.

Overview

The tool enables a semi-automated workflow to extract and load CAD data from an Asset Administration Shell (AASX) into Siemens NX.

Key Features:

- Interactive selection of an AASX file
- User-defined selection of a specific STEP model via part name
- Targeted extraction of STEP files instead of generic search
- Automatic conversion from STEP → NX PRT format
- Automatic insertion of components into an assembly
- Support for positioning multiple instances of the same model
- Logging and error output via NX Listing Window

Basic functionality

The workflow has been extended to allow **targeted selection of CAD models** and **structured import into NX assemblies**.

Process:

1. **Start of the NX script**
   The user launches the NX script inside Siemens NX.
2. **Execution of external AAS script**
   The NX script starts the external Python script via subprocess.
3. **Selection of AASX file**
   A file dialog is opened, allowing the user to select an `.aasx` file.
4. **Input of part name**
   The user is prompted to enter a specific part name
   (e.g., `GPLE60-3S`).
5. **Targeted STEP extraction**
   The script:
   - Parses the AASX file
   - Searches for STEP models
   - Compares filenames with the user input
   - Extracts only the matching STEP file
6. **Temporary storage**
   The STEP file is stored locally.
7. **Import into NX**
   The NX script:
   - Converts the STEP file into a `.prt` file
   - Inserts it as a component into the current assembly

- Optionally inserts multiple instances with positioning

Code structure and individual functions

The code consists of two scripts:

- **NX launcher script** (performs the import in NX)
- **External AAS script** (extracts the STEP model from the AASX)

NX launcher script

The NX script has been extended to support:

- STEP → PRT conversion
- Component-based assembly building
- Multiple component placement

`main()`

The entry point of the NX script.

Functions:

- Opens the NX Listing Window to display status messages
- Starts the external Python script via subprocess
- Checks whether the STEP file was created
- Ensures that an active NX part exists
- Imports the STEP file into NX

`ensure_work_part()`

This function ensures that an active NX part exists.

If no work part is available:

- a new part is created
- any existing old part is deleted
- the new part is set as the work part

This prevents NXOpen functions from being executed without a valid target part.

`import_step_to_prt()`

This function converts a STEP file into an NX `.prt` file.

Features:

- Uses NX Step242Importer
- Creates a new NX-native part file
- Enables:
  - Solids
  - Surfaces
  - Curves
  - PMI data
- Outputs a `.prt` file in the same directory

This replaces direct STEP import into the work part.

`add_prt_as_component()`

This function inserts a `.prt` file into the current assembly.

Features:

- Adds part as a component
- Supports positioning via:
  - X, Y, Z coordinates
- Assigns a component name
- Uses NX ComponentAssembly API

`import_step_into_nx_as_component()`

This is the central import function.

Process:

1. Ensure active work part exists
2. Convert STEP → PRT
3. Insert PRT as component into assembly
4. Apply position and naming

This replaces the old direct STEP import.

Multiple component placement

The script supports inserting multiple instances of the same STEP model:

Example:

- Component 1 at (0, 0, 0)
- Component 2 at (150, 0, 0)

This enables simple assembly creation directly from AAS data.

External script (AAS_TO_NX.py)

This script extracts a STEP model from an AASX file.

`main()`

The entry point of the external script.

Process:

1. Opens a file selection dialog
2. The user selects an .aasx file
3. The file is analyzed
4. A STEP model is extracted
5. The STEP file is saved locally

`select_aasx_file()`

This function opens a file dialog using Tkinter.

The user can select an .aasx file.

If no file is selected, the process is terminated.

`extract_step_from_aasx_by_name()`

This function performs a **targeted search** for a STEP file inside the AASX.

Process:

1. Import AASX file
2. Iterate through all AAS shells
3. Extract models from each AAS
4. Check metadata for STEP format
5. Extract filename (without extension)
6. Compare with user-defined part name
7. If match is found:
    - Extract file from file store
    - Save to temporary location
8. Return path to STEP file

If no matching STEP file is found:

- Function returns `None`

`ask_part_name()`

This function prompts the user to enter a part name.

- Uses Tkinter dialog
- Input is validated (no empty values)
- Defines which STEP model should be extracted

This replaces the previous "take first STEP file" logic.

Installation & execution in NX

1. Ensure that Python is installed.
2. Ensure that the NX Python API is available.
3. Place both scripts in your project directory.

Required scripts:

- NX launcher script:
  AAS_TO_NX.py

4. Open Siemens NX.
5. Start the NX script

The script will then automatically start the complete workflow.

Troubleshooting

STEP file not found

Possible causes:

- No STEP file matches the entered part name
- Naming mismatch between AASX and user input

External script aborted

Return codes:

- `1` → No file selected
- `2` → No matching STEP found

Import into NX fails

Possible causes:

- STEP → PRT conversion failed
- NX importer configuration issue
- File system access problem

Possible improvements

- Selection of part names from dropdown instead of manual input
- Preview of available STEP models inside AASX
- Support for multiple different components from one AASX
- Automatic placement based on metadata (position/orientation)
- Configurable paths instead of hardcoded values
- Logging to file instead of only NX Listing Window

NX addPartID for AAS

This project includes a Siemens NX Open application (C#) created using the NX Block UI Styler. For detailed instructions and technical background on working with this tool, please refer to the official Siemens manual **"NX Open Programmer's Guide"**, specifically the chapters on the **"Block UI Styler Introduction"** (for dialog creation and callbacks).

The tool enables designers to assign a unique ID (Part ID) to an active NX part via a simple graphical user interface. This ID is stored as a part attribute (`PART_ID`) and is used to link the 3D model with the **Asset Administration Shell (AAS)** for digital twins.

Basic functionality (Adding the Part ID for the AAS)

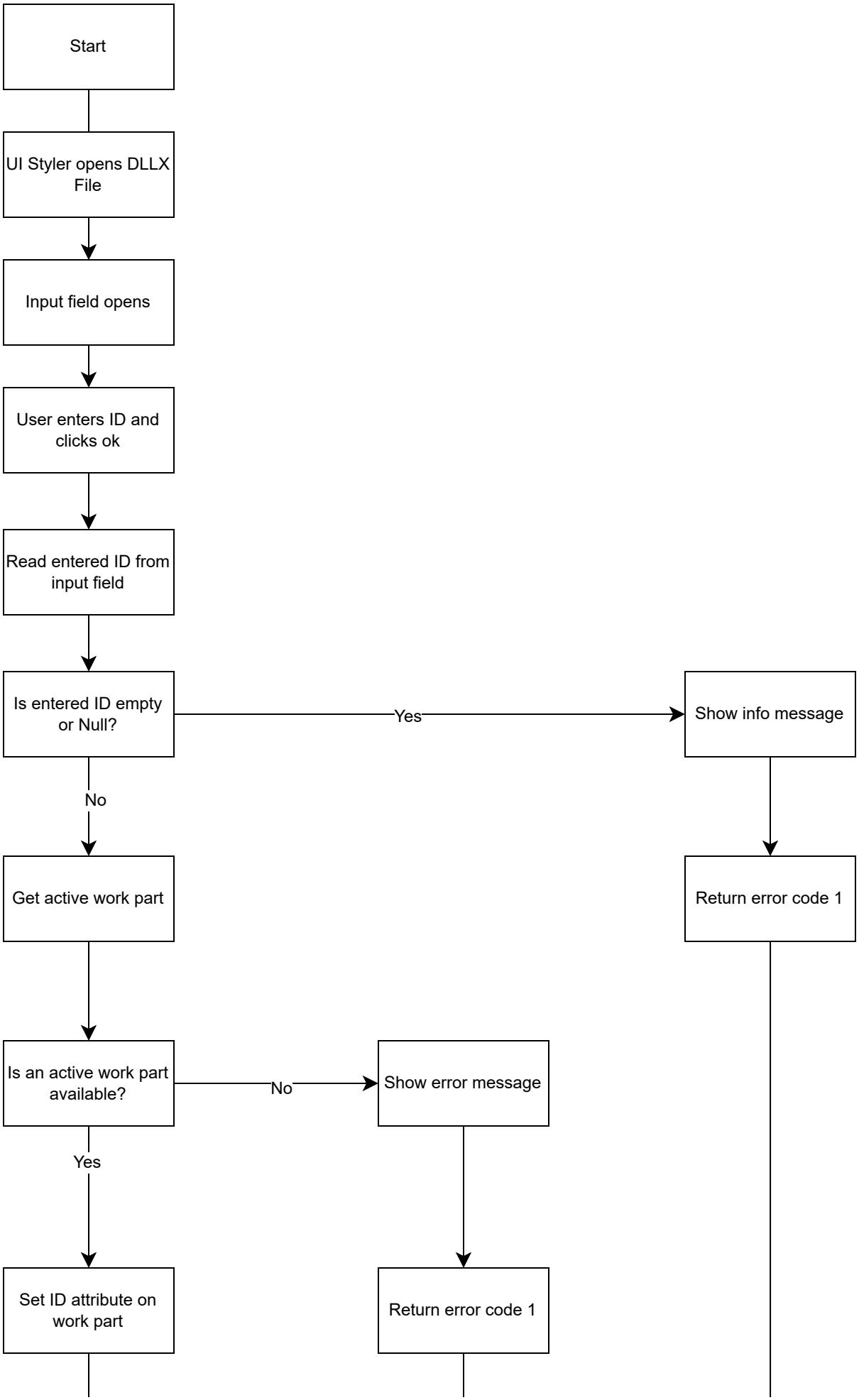
To make a 3D model usable in Industry 4.0 scenarios, it must be uniquely identifiable. This script addresses this requirement by providing a defined workflow in NX:

1. **User input:** The user starts the script in NX. A dialog box (defined by the `.dlx` file) opens, in which the desired ID for the Asset Administration Shell is entered.
2. **Validation:** The tool verifies whether an ID has been entered and whether a part is opened and active (Work Part) in NX.
3. **Attribute assignment:** The entered string is written directly into the part metadata via the NX Open API. The attribute **`PART_ID`** is created or overwritten.

Once the part is saved and, for example, exported as a STEP file (including metadata), the AAS can read this `PART_ID` and associate the part with a digital twin.

Technical workflow diagram

The following SVG file shows a **flowchart** that describes the technical functionality of the implemented process.


Start
UI Styler opens DLLX File
Input field opens
User enters ID and clicks ok
Read entered ID from input field
Is entered ID empty or Null?
Yes
Show info message
Return error code 1
No
Get active work part
Is an active work part available?
No
Show error message
Return error code 1
Yes
Set ID attribute on work part

Code structure and the individual functions

The code is based on the standard template of the NX Block UI Styler. Below is a detailed description of the key functions in `add_PartID.cs` :

`Main()`

The entry point of the application. A new instance of the `add_PartID` class is created, and the dialog is displayed on the user's screen via the `Show()` method.

`initialize_cb()`

(Initialize Callback)
This method is executed immediately when the dialog is loaded. It establishes the connection between the C# logic and the graphical user interface (the `.dlx` file). Here, the input field (block type: String) is located in the UI and assigned to the variable `string0` , so it can be accessed later in the process.

`apply_cb()`

(Apply Callback)
This function is triggered when the user clicks **"Apply"**:

- **Read values:** Retrieves the text from the input field (`string0.Value`).
- **Validation checks (If-Statements):**
  - Checks whether the field is empty. If so, an informational message is displayed and the process is stopped.
  - Checks whether an active work part (`workPart`) exists. Without a part, no attribute can be assigned.
- **Write operation:** The command `workPart.SetAttribute("PART_ID", eingegebeneID);` writes the actual attribute to the NX part.
- **Session Undo:** Creates a visible undo step ("Part ID set"), to keep the NX session clean.

`ok_cb()`

(OK Callback)
Executed when the user clicks the "OK" button. This function internally calls `apply_cb()`, thereby performing the attribute assignment and then closing the dialog window.

`GetUnloadOption()`

Defines how the program is handled in NX memory after execution. Here, `Session.LibraryUnloadOption.Immediately` is set. This means that memory is released immediately after the script has been executed. This prevents memory leaks during an NX session.

The User Interface (`add_PartID.dlx`)

The included XML file (`.dlx`) defines the visual appearance of the NX dialog. It includes:

- The window title: **"Add Part ID"**
- A group (`group0`) for visual structuring
- The essential text input field (`string0`) labeled **"Enter Part ID"**.

Installation & Execution in NX

1. Download the files `add_PartID.cs` and `add_PartID.dlx`
2. Ensure that both files are located in the same directory or placed according to the NX environment variables (e.g., in the `application` folder under `UGII_USER_DIR`)
3. Open a part in Siemens NX.
4. Execute the C# file via (File -> Execute -> NX Open) or integrate it as a button into your NX user interface (Ribbon).

Troubleshooting

- **Error during startup (dialog cannot be found):** By default, the C# script automatically locates the `add_PartID.dlx` file within the same directory where the script is stored. There is no need to

manually adjust the path in the code, as long as the .cs file and the .dlx file remain in the same folder. Alternatively, the `.dlx file` can be placed in a standardized NX directory `($UGII_USER_DIR/application/)` for automatic detection by NX.".

- **Message "No active part available":** The script was executed without a part being open, or the displayed part is not the "Work Part". Set the part as the Work Part by right-clicking it in the Assembly Navigator and try again.

05 NX Sync Properties with AAS

Text

06 Attribute Export Skript

This Python script uses the NXOpen Python API to read specific attributes of an open NX part and export them into a CSV file.

The CSV file contains selected mass properties of a part, e.g.:

- Volume
- Mass
- Surface area
- Material

The file is automatically saved in the same directory as the NX part.

Basic Functionality

The script executes the following process:

1. Establish a connection to the current NX session
2. Access the active work part

3. Define the attributes to be read
4. Iterate over all attributes of the part
5. Extract the required values
6. Write the data to a CSV file

Code Structure and Functions

main()

The entry point of the script.

Functions:

- Initialize Session: Access the current `NXOpen.Session`.
- Get Active Part: Access the `workPart` (abort if none is open).
- Determine Script Directory: Use `Path(__file__)` to get the folder where the script is stored.
- Define Target Attributes: Specify required keys (e.g., `MassPropMass`, `Material`).
- Update Mass Properties: Use `MassCalculationBuilder` to ensure geometry data is up-to-date.
- Collect Attribute Values: Iterate through all part attributes and filter by defined keys.
- Define CSV Path: Combine the script directory with the part name for the export file.
- Export to CSV: Write the filtered values into a `.csv` file located in the script's folder.

Definition of the required attributes

The following attributes are exported:

```
wanted_attributes = [
"MassPropVolume",
"MassPropMass",
"MassPropArea",
"Material"
]
```

These attributes correspond to mass properties of an NX part.

Attribut	Description
MassPropVolume	Volume of the part
MassPropMass	Mass of the part
MassPropArea	Surface area of the part
Material	Material of the part

Creation of the export file path

The file path of the CSV file is generated automatically

Installation & Execution in NX

Ensure that Siemens NX with the NX Open Python API is installed. Save the script in an NX script directory or a project folder. Open a part in Siemens NX. Execute the script via the following menu:

- File → Execute → NX Open
- Select the Python script
- After execution, a CSV file is automatically generated

Troubleshooting (Fehlerbehebung)

No CSV file is generated
Possible causes:

- No active work part is available
- The part does not contain the required attributes
- Missing write permissions in the target directory (CSV File already open)

Attributes are not found
Possible causes:

- Attributes have different names
- Attributes are not stored as user attributes

- Alias and title differ

07 Recycling NX R-BOM to AAS

Toolbox in NX

The connection of the CAD system Siemens NX to the Asset Administration Shell (AAS) enables a continuous, bidirectional flow of information. This transforms the static 3D model into an integral component of the Digital Twin. Using the NX Open API, a wide variety of functions can be implemented, expanding the functional scope for AAS use cases almost limitlessly.

The following core functions and workflows are made possible by this integration:

1. Automated import of CAD data from the AAS

- **AASX extraction:** CAD models (such as STEP files) referenced within an AASX container can be specifically searched for, extracted, and saved in a temporary directory using external Python scripts.
- **Automatic loading in NX:** A launcher script executed in NX picks up this data and automatically imports it into the work environment via the NX Step Importer.
- **Assemblies and single parts:** Complex assembly structures as well as individual parts can be loaded. *Note on positioning: Via the NX Open API, exact transformation matrices (coordinates) can be passed to the imported bodies during loading or within the assembly.*

2. Assignment of a unique Asset ID (Digital Thread)

- **ID linking:** To uniquely assign a CAD part to a specific AAS instance, a unique ID can be assigned to the model directly in NX.
- **Attribution:** This assignment is done by writing metadata into the part attributes (e.g., creating the attribute `PART_ID`). If the part is exported later, this ID is retained as meta-information within the model and can be read by Industry 4.0 systems.

3. Data export for Bills of Materials (BOM) and AAS synchronization

- **Reading properties:** Specific physical mass properties of an NX part (such as volume, mass, and surface area) can be automatically read out using a Python script.
- **BOM creation:** These extracted CAD attributes can be exported (e.g., as a CSV file). This data forms the basis for detailed bills of materials (such as an R-BOM for recycling scenarios), which can then be linked directly into the structure of the AASX file.

4. UI customization and user interaction

- **NX Block UI Styler:** Native graphical user interfaces (dialog boxes) can be created directly in NX. For example, the user can enter IDs via a convenient text field while validation checks run in the background (e.g., whether an active part is open).
- **Seamless integration:** The NX user interface can be customized by adding scripts as interactive buttons directly to the NX menu bar (ribbon). This ensures a high level of user-friendliness.
- **Limitless extensibility:** Since NX can act both as a launcher for external Python scripts and execute internal C#/Python logic, there are virtually no technical limitations for automation and data linking with the Digital Twin.

Comparison of CAD Systems in the Context of the Asset Administration Shell (AAS)

If the question arises during development as to which CAD system should be used, practical experience from various projects can be drawn upon. As shown in the following table, the three established CAD systems Siemens NX, PTC Creo, and Autodesk Fusion 360 differ from each other, particularly in terms of licenses, functional scope, and user-friendliness.

Criterion	Siemens NX	PTC Creo	Autodesk Fusion 360
Licenses	0 Enterprise license required to work with journals	+/-/0	🚧 Work in Progress

Criterion	Siemens NX	PTC Creo	Autodesk Fusion 360
Functional Scope	+ Extensive, everything that can be covered via Python	+/-/0	🚧 Work in Progress
User-Friendliness	+ Through integration into the NX interface	+/-/0	🚧 Work in Progress

Table: Comparison of selected CAD systems for AAS integration

Prerequisites

For the successful setup and execution of the integration between Siemens NX and the AAS, the following system requirements must be met:

Software & Licenses

- ☐ **Siemens NX:** Local installation (e.g., version 2021).
- ☐ **NX License:** Enterprise or NX Open license for executing and compiling scripts and external programs.
- ☐ **AASX Package Explorer:** Tool for viewing and validating the Asset Administration Shells.

Development Environment & Languages

- ☐ **Python:** Standalone installation (version 3.11 or higher) for adapters, because the Python version in NX 2021 is only version 3.8 and does not include the lxml library.
- ☐ **IDE:** Visual Studio (e.g., for C#) or Visual Studio Code (for Python) to customize the scripts.

System Rights & Configuration

- ☐ **Administrator Rights:** Local Windows admin rights (for installations via `pip` and setting environment variables).
- ☐ **NX Role:** An advanced user role (e.g., "Advanced") to unlock the developer tab in the NX user interface.

Integration of the developed scripts into the Siemens NX CAD environment

Integration of Python Scripts in Siemens NX

There are multiple ways to integrate and execute Python scripts in Siemens NX.

Option 1: Execution via Journal

One approach is to execute the script using the **Journal playback functionality**.

Steps:

- Add the *Journal Playback* command in NX
- Select the desired Python script
- Execute the script directly within the NX environment

Option 2: User-defined command

Another approach is to create a **custom user command** in NX.

- A new user command is defined
- The Python script is assigned as the corresponding action
- The script can then be executed like a native NX command

Important note

It is important to note that scripts are referenced via **file paths** in Siemens NX.

- If the script is moved or renamed,
- the corresponding path in NX must be updated

Otherwise, the script can no longer be executed.

Integration of C# based UI Styler Scripts

Scripts created with the **NX UI Styler** can also be executed via the **Journal Replay** functionality.

However, there are some important aspects to consider:

- UI Styler-based scripts require both a `.cs` file and a corresponding `.dllx` file.
- These files must be correctly linked to each other.
- The `.cs` file contains a file path reference to the associated `.dllx` file.
- This path must be updated before the first execution to ensure correct functionality.

Additional Integration Option

UI Styler scripts can also be integrated as a **custom user command**:

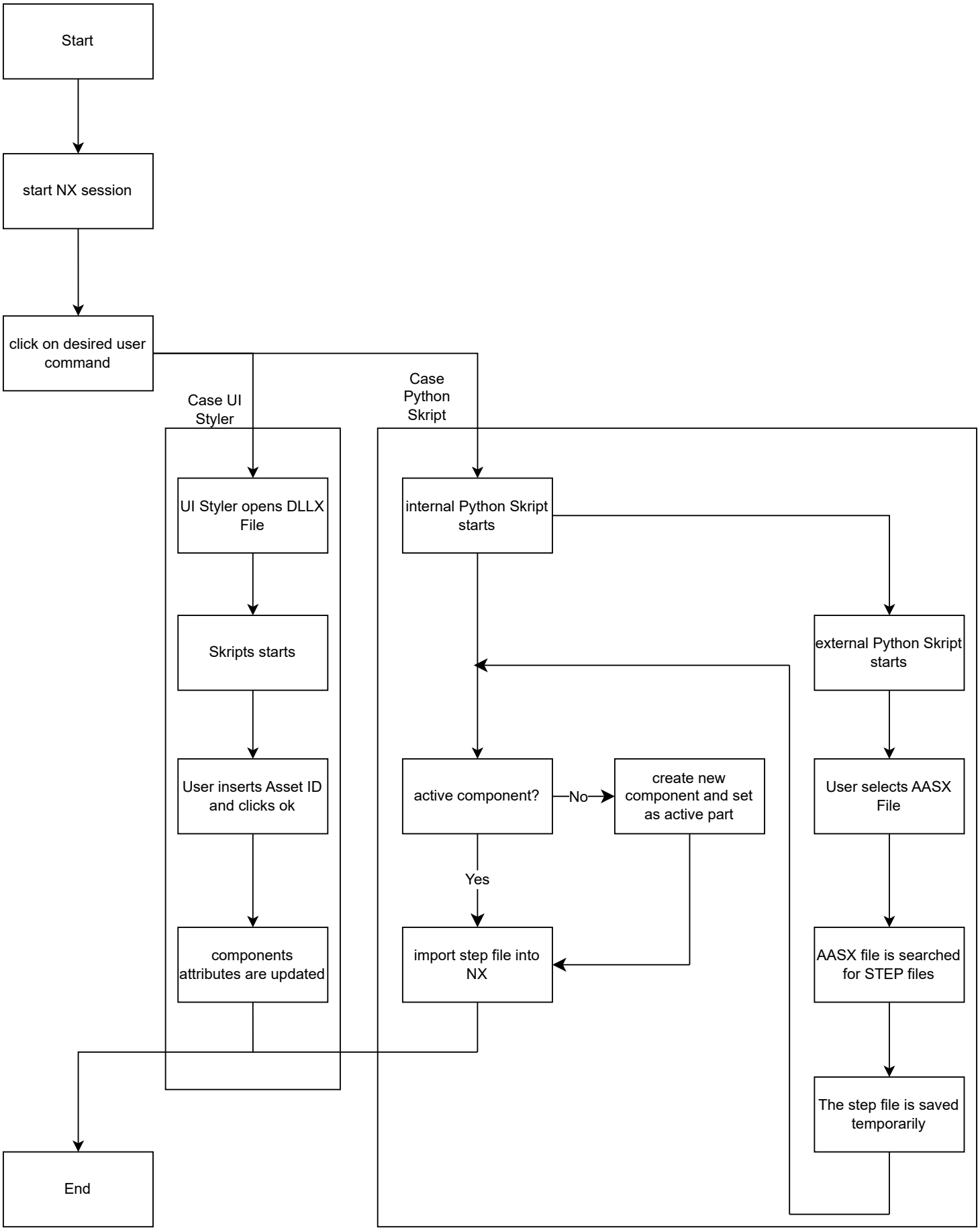
- The `.cs` script is assigned as the execution action.
- The command can then be added to the NX user interface (e.g., toolbar or menu).

This allows seamless integration into regular NX workflows.

Workflow Diagram

The following flowchart presents the internal process flow of the interaction between a UI Styler-based interface and a Python script.

The UI Styler component is exemplified by the `add_PartID` function, whereas the Python-based backend logic is represented by the `NX AASX to STEP Import Tool` function.


Start
start NX session
click on desired user command
Case UI Styler
Case Python Skript
UI Styler opens DLLX File
Skripts starts
User inserts Asset ID and clicks ok
components attributes are updated
internal Python Skript starts
active component?
No
create new component and set as active part
Yes
import step file into NX
external Python Skript starts
User selects AASX File
AASX file is searched for STEP files
The step file is saved temporarily
End

Conclusion and Outlook

Conclusion

This project has successfully built a functioning bridge between the standardized Asset Administration Shell (AAS) and the CAD system Siemens NX. In doing so, it has shown that a direct, bidirectional connection between both worlds is essential. Data must flow seamlessly throughout the entire product lifecycle – whether it is generated in CAD and transferred to the AAS, or vice versa. A clear practical example of this is the automated calculation of the material weights of an assembly in CAD, which are then made directly available for processes in the AAS.

However, there were some hurdles to overcome at the beginning of the project. One challenge was the licensing model of Siemens NX: the "NX Open" functions, which are absolutely necessary for interface programming, are missing in the free student version.

Nevertheless, we were able to successfully implement the essential tools for automated data exchange within the scope of the project:

- **Import of 3D data from the AAS:** A tool specifically extracts STEP models from AASX containers, converts them into native NX PRT files, and automatically inserts them into NX, including the option for positioning.
- **Assignment of an Asset ID to a part ("addPartID"):** Using an integrated user interface, designers can assign a unique Asset ID (`PART_ID`) to their models directly in NX. This linking of the 3D model with its AAS instance is a fundamental prerequisite for Industry 4.0.
- **Synchronize properties & Data export from CAD:** A Python script automatically reads physical properties (volume, mass, material) from NX and exports them as a CSV file – the ideal basis for recycling bills of materials (R-BOMs).
- **Seamless UI integration:** All developed functions were integrated as buttons directly into the NX menu bar (ribbon). They do not disrupt the normal design workflow and are highly intuitive to use.

Outlook

The created interfaces form a solid foundation. To completely close the data loop, the following further developments are recommended for future projects:

- **Web-based search function:** The goal is to use an Asset ID in the browser to directly download the appropriate AASX file, automatically find the contained CAD model, and seamlessly open it in NX.

- **Direct AASX export:** The current workaround using CSV files should be eliminated. In the future, material and geometry data from the CAD system should be written directly into the standardized AASX format so that it can be used in the AAS without intermediate steps.

Once this bidirectional workflow is fully established in Siemens NX, our methodology can serve as a blueprint. The concepts developed here can then be easily transferred to other CAD systems to achieve true, cross-system interoperability.

Tools and Sources

Tools Used

The following tools supported the implementation and documentation of the project:

- **DeepLcom:** For translations from German into English.
- **Gemini:** As an AI assistant for Markdown formatting and code snippets.
- **Drawio:** For creating the diagrams.
- **Visual Studio:** As the development environment for programming.
- **GitHub:** For version control and publishing the code.
- **Siemens NX:** For the CAD work.

Sources

The theoretical and technical basis of this project relies on the following documents:

[1] Industrial Digital Twin Association (IDTA), *Specification of the Asset Administration Shell - Part 1: Metamodel*, Version 3.0. Frankfurt am Main, Germany: IDTA e.V., 2023. [Online]. Available: https://industrialdigitaltwin.org/wp-content/uploads/2023/06/IDTA-01001-3-0_SpecificationAssetAdministrationShell_Part1_Metamodel.pdf

[2] European Commission, "Commission Regulation (EU) 2019/2022 of 1 October 2019 laying down ecodesign requirements for household dishwashers," *Official Journal of the European Union*, L 315, pp. 267-284, Dec. 2019.

[3] European Parliament and Council of the European Union, "Regulation (EU) 2024/1781 of 13 June 2024 establishing a framework for the setting of ecodesign requirements for sustainable products," *Official Journal of the European Union*, L, Jun. 2024.

[4] European Parliament and Council of the European Union, "Regulation (EU) 2023/1542 of 12 July 2023 concerning batteries and waste batteries," *Official Journal of the European Union*, L 191, pp. 1-117, Jul. 2023.

[5] European Parliament and Council of the European Union, "Directive 2012/19/EU of 4 July 2012 on waste electrical and electronic equipment (WEEE)," *Official Journal of the European Union*, L 197, pp. 38-71, Jul. 2012.

[6] European Parliament and Council of the European Union, "Directive 2008/98/EC of 19 November 2008 on waste and repealing certain Directives," *Official Journal of the European Union*, L 312, pp. 3-30, Nov. 2008.

[7] European Parliament and Council of the European Union, "Regulation (EC) No 1907/2006 of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH)," *Official Journal of the European Union*, L 396, pp. 1-849, Dec. 2006.

[8] Siemens Digital Industries Software, *NX Open Programmer's Guide: Block UI Styler and Journaling*, Siemens PLM Software Inc., Plano, TX, USA. [Online]. Available at: https://support.sw.siemens.com/

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Siemens NX 2021 UI-Styler

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